Validating Transformer-Based Financial Encoders on Structured XBRL Data: A Benchmark Study

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Abstract

We evaluate the FT-Transformer architecture as a standalone encoder for structured financial data extracted from SEC EDGAR XBRL filings, benchmarking it against logistic regression, XGBoost, and gradient-boosted trees on four financial distress prediction tasks. Using a strictly temporal train/test split on 33,737 company-year observations (2012–2023), and giving the FT-Transformer and the baselines an equal Optuna tuning budget selected on the same expanding-window temporal cross-validation, we find that the FT-Transformer meets the AUROC gate (3/4 wins): it wins on earnings restatement, bankruptcy, and SEC enforcement, with stock decline a sub-threshold near-tie. (Against default-hyperparameter XGBoost the margin is wider—directionally positive on all four outcomes, +0.015 to +0.075 AUROC, significant on two by paired bootstrap—but we treat the equal-budget comparison as decisive.) Expanding-window walk-forward validation (each fold's preprocessing fit on its own training split) across the COVID-19 boundary shows this advantage is temporally stable for earnings restatement (the FT-Transformer wins all seven folds) and largely robust for bankruptcy (6/7 folds, losing only the 2020–2021 window). The FT-Transformer also retains calibration advantages on the highest-prevalence outcome and produces more informative probability distributions on rare-event outcomes. Self-supervised pretraining via masked feature reconstruction provides additional lift, particularly on low-base-rate outcomes (+0.092 AUROC on SEC enforcement actions at mask ratio 0.50). Multi-seed evaluation (3 seeds) shows modest variance on three outcomes (std ≤ 0.013) but substantial variance on the rare SEC-enforcement outcome (std = 0.044). Together with these calibration and representation-learning properties, the met gate supports adopting the FT-Transformer as the base encoder for a larger research program investigating cross-modal joint-embedding predictive architectures for financial risk signals. All data extraction pipelines, feature definitions, and training code are released as open-source software.

1 Introduction

Deep learning has achieved strong results across many domains, yet its adoption for tabular financial data remains limited. Tree-based ensembles—particularly XGBoost [Chen and Guestrin, 2016] and gradient-boosted trees—continue to dominate applied financial modeling, and recent comprehensive benchmarks confirm that they remain competitive with or superior to deep learning on many tabular tasks [Grinsztajn et al., 2022, Shwartz-Ziv and Armon, 2022]. This presents a problem for research programs that aim to build multimodal financial architectures: if the base encoder for structured financial data cannot match conventional baselines, downstream gains from cross-modal integration are difficult to attribute.

The FT-Transformer [Gorishniy et al., 2021] addresses several known failure modes of deep learning on tabular data. By tokenizing each input feature through a learned linear projection and processing the resulting sequence with standard Transformer attention, it avoids the feature-interaction bottlenecks of MLPs while preserving the inductive biases that make attention effective for heterogeneous input modalities. Gorishniy et al. [2021] demonstrated that the FT-Transformer matches or exceeds gradient-boosted decision trees on a range of tabular benchmarks, making it a plausible backbone for financial data.

However, financial tabular data presents distinct challenges that generic tabular benchmarks do not capture. XBRL (eXtensible Business Reporting Language) filings contain features with extreme tail distributions, systematic missingness driven by reporting standards rather than randomness, and temporal dependence structures that violate the i.i.d. assumptions of standard cross-validation. Whether the FT-Transformer's advantages generalize to this setting is an open empirical question.

Self-supervised pretraining offers an additional potential advantage. In the natural language and computer vision domains, pretraining on unlabeled data before fine-tuning on task-specific labels consistently improves performance, especially in low-label regimes [Devlin et al., 2019, He et al., 2022]. For financial data, where distress events are rare and ground-truth labels are expensive to obtain, even modest gains from self-supervised initialization could be practically significant. We investigate masked feature reconstruction—a tabular analog of masked language modeling—as a pretraining strategy for the FT-Transformer.

This paper makes the following contributions:

1. We construct a reproducible dataset of 16 XBRL-derived financial features for the full universe of SEC 10-K filers from 2012 to 2024, with four binary distress outcome labels, using only publicly available data sources.

2. We benchmark the FT-Transformer—with hyperparameters selected via a 36-configuration grid search—against logistic regression (with both traditional ratios and full feature sets), XGBoost, and a histogram-based gradient-boosted tree on distress prediction, using a strictly temporal evaluation protocol that prevents look-ahead bias.

3. We provide a rigorous like-for-like tuned comparison in which both the baselines and the FT-Transformer are Optuna-tuned (30 trials per model, 3-fold expanding-window temporal CV), alongside default-hyperparameter baselines, and report multi-seed variance (3 seeds) for the FT-Transformer.

4. We assess the temporal stability of the FT-Transformer advantage with expanding-window walk-forward validation spanning the COVID-19 boundary (seven folds, test windows 2016–2017 through 2022–2023).

5. We evaluate masked feature reconstruction as a self-supervised pretraining objective for the FT-Transformer and measure its effect on downstream classification performance.

6. We release the complete data extraction pipeline, feature engineering code, model implementations, and evaluation scripts as open-source software to support reproducibility and further research.

This work constitutes Study 0 of the Fin-JEPA research program, a gated sequence of studies that progressively builds toward a cross-modal joint-embedding predictive architecture for financial risk signals [Assran et al., 2023]. Giving both sides an equal Optuna tuning budget on the same temporal cross-validation, the FT-Transformer meets the strict AUROC gate (3/4 wins); against default-hyperparameter XGBoost the margin is wider still (directionally positive on all four outcomes, significant on two). Walk-forward validation shows the advantage is temporally stable on the highest-signal outcome (earnings restatement, all seven folds) and largely robust on bankruptcy (six of seven), with the rare SEC-enforcement outcome the least reliable. Together with its calibration properties, its suitability as a pretrained representation backbone, and the gains from self-supervised pretraining, this supports adopting it as the structured financial modality encoder in subsequent studies.

2 Data

2.1 Company Universe

We construct a survivorship-bias-free universe of all entities that filed SEC Form 10-K annual reports between fiscal years 2012 and 2024. The universe is derived from two EDGAR data sources: (i) the quarterly full-index files, which enumerate all filings by type and date, and (ii) the per-company submissions API,[1] which provides metadata including ticker symbols, exchange listings, SIC codes, and fiscal year end dates.

The 2012 start date is chosen because XBRL tagging became mandatory for all SEC filers in that year, ensuring consistent structured data availability across the universe. The universe includes all entities regardless of current listing status—delisted, bankrupt, and acquired companies are retained to avoid survivorship bias. We assign each company to one of 12 sectors using the Fama–French industry classification mapped from 4-digit SIC codes.

The resulting universe contains approximately 8,000 unique companies. Each company-year observation is identified by its CIK (Central Index Key), ticker symbol, fiscal year, fiscal period end date, and 10-K filing date.

2.2 XBRL Feature Extraction

We extract 16 financial features per company-year from the EDGAR Company Facts API,[2] which provides structured XBRL data tagged with US-GAAP taxonomy concepts. For each target feature, we define an ordered list of candidate XBRL concept tags (e.g., `us-gaap:Assets`, `us-gaap:AssetsCurrent`) and resolve to the first available tag, accommodating variation in reporting practices across companies.

We restrict extraction to annual (10-K / FY) filings. When multiple filings exist for the same company-year (e.g., due to amendments), we deduplicate by retaining the most recently filed version. All HTTP responses are cached to disk per CIK, and requests are rate-limited to $\leq$10 per second in compliance with SEC EDGAR fair-use policies.

The 16 raw features span three financial statement categories:

Balance sheet (8 features). Total assets, total liabilities, stockholders' equity, current assets, current liabilities, retained earnings, cash and cash equivalents, and total debt (computed as the sum of long-term and short-term debt when reported separately).

Income statement (5 features). Total revenue, cost of goods sold, operating income, net income, and interest expense.

Cash flow statement (3 features). Cash from operations, cash from investing activities, and cash from financing activities.

2.3 Feature Engineering

From the 16 raw XBRL features, we derive two additional feature sets:

Financial ratios (12 features). We compute standard financial ratios spanning four categories: leverage (debt-to-equity, debt-to-assets, interest coverage, CFO-to-debt), liquidity (current ratio, quick ratio), profitability (return on assets, return on equity, gross margin, operating margin, net profit margin), and distress (Altman Z-score). The Altman [1968] Z-score is

[1] `https://data.sec.gov/submissions/CIK\{cik\}.json`

[2] `https://data.sec.gov/api/xbrl/companyfacts/CIK\{cik\}.json`

computed using the original 5-component formula:

$$Z = 1.2\,X_1 + 1.4\,X_2 + 3.3\,X_3 + 0.6\,X_4 + 1.0\,X_5 \tag{1}$$

where

X_1 = working capital/total assets, X_2 = retained earnings/total assets,
X_3 = operating income/total assets, X_4 = equity/total liabilities,
X_5 = revenue/total assets.

All divisions use a minimum absolute denominator threshold of 10^{-8} to avoid numerical instability, returning NaN where the denominator is effectively zero.

Year-over-year changes (16 features). For each of the 16 raw XBRL features, we compute the year-over-year percentage change as $(x_t - x_{t-1})/|x_{t-1}|$. Using the absolute value of the prior-year value in the denominator handles sign changes (e.g., a firm moving from negative to positive net income). The first observation per company receives NaN by construction and is flagged with `is_first_year` = 1.

Missingness indicators. For each numeric feature retained after coverage pruning, we add a binary indicator variable equal to 1 if the original value was missing and 0 otherwise. These flags are not normalized.

The default configuration uses all feature groups (raw + ratios + YoY + missingness flags). Before coverage pruning, this yields 45 numeric features (16 raw + 12 ratios + 16 YoY + 1 `is_first_year`) plus one categorical feature (`sector_idx`, Fama–French 12-industry). After pruning features with less than 50% non-null coverage in the training split and appending per-feature binary missingness indicators, the final model input comprises 83 continuous features and 1 categorical feature.

2.4 Feature Normalization

Financial features exhibit extreme tail behavior that renders standard z-score normalization unreliable. We apply a three-stage normalization pipeline, fit exclusively on the training split to prevent look-ahead bias:

1. **Median imputation.** Missing values are replaced with the per-feature median computed on the training split.
2. **Winsorization.** Values are clipped to the 1st and 99th percentile bounds computed on the training split, attenuating the influence of extreme outliers.
3. **Quantile transformation.** We apply scikit-learn's `QuantileTransformer` with `output_distribution='normal'` and up to 1,000 quantiles (minimum 2), which maps the empirical rank distribution of each feature to a standard normal distribution. This is the default normalization method; z-score standardization is available as an alternative.

Features with less than 50% non-null coverage in the training split are pruned before normalization. This threshold is applied at the training-split level to prevent information leakage from the validation or test periods.

2.5 Market Data

We collect daily adjusted OHLCV (open, high, low, close, volume) price data via yfinance for all tickers in the company universe, covering the period 2010-01-01 through 2024-12-31. The 2010 start date provides a two-year price history buffer before the first fiscal year in our sample (2012).

For each company-year observation, we align market data to the 10-K filing date and compute forward returns at four horizons: 21, 63, 126, and 252 trading days. Returns are market-adjusted by subtracting the corresponding S&P 500 (^GSPC) return over the same window, and sector-adjusted using Fama–French 12-industry sector ETF returns. We compute 63-day average daily volume preceding the filing date as an additional feature. Companies with insufficient post-filing price data (indicating delisting) are flagged.

2.6 Distress Labels

We construct four binary distress outcome labels per company-year, covering distinct dimensions of financial distress. Labels are assigned based on events occurring within a 365-day horizon following the fiscal period end date. Label columns use nullable Int8 dtype, where 0 indicates no event, 1 indicates a distress event, and NaN indicates that the label is unavailable for that observation (not equivalent to a negative label).

Table 1: Distress outcome definitions.

Outcome	Source	Definition
`stock_decline`	Market data (computed)	Market-adj. 252-day return $< -20\%$
`earnings_restate`	EDGAR filing index	10-K/A amendment filed within horizon
`sec_enforcement`	Dechow et al. AAER database	SEC enforcement release within horizon
`bankruptcy`	UCLA-LoPucki BRD / Compustat	Chapter 7 or 11 filing within horizon

The first two outcomes (`stock_decline` and `earnings_restate`) are derived entirely from publicly available EDGAR data. The remaining two require external datasets: the Dechow et al. AAER database for SEC enforcement actions and the UCLA-LoPucki Bankruptcy Research Database for bankruptcy filings. When external label files are not present, the corresponding label columns are all-NaN and the outcome is excluded from evaluation. Delisted companies with insufficient post-filing price data are treated as stock decline events.

On the test split, approximate positive/negative counts are: `stock_decline` 1,200/1,300, `earnings_restate` 450/4,050, `sec_enforcement` 40/3,960, and `bankruptcy` 45/3,955.

3 Method

3.1 FT-Transformer

We adopt the FT-Transformer architecture of Gorishniy et al. [2021] as our primary encoder for structured financial data. The architecture consists of three components:

Feature Tokenizer. Each scalar input feature x_i is projected to a d-dimensional token embedding via a per-feature affine transformation: $\mathbf{t}_i = \mathbf{W}_i x_i + \mathbf{b}_i$, where $\mathbf{W}_i \in \mathbb{R}^d$ and $\mathbf{b}_i \in \mathbb{R}^d$ are learnable parameters. Feature weights are initialized with Kaiming uniform initialization ($a = \sqrt{5}$). A learnable [CLS] token $\mathbf{t}_{\text{cls}} \in \mathbb{R}^d$ is prepended to the token sequence, yielding a sequence of $n + 1$ tokens where n is the number of input features.

Transformer Encoder. The token sequence is processed by a stack of L standard Transformer encoder layers [Vaswani et al., 2017]. Each layer applies multi-head self-attention followed by a position-wise feed-forward network, with pre-layer normalization (norm-first configuration). The feed-forward hidden dimension is set to $d \times f$ where f is a configurable expansion factor. We use PyTorch's `nn.TransformerEncoderLayer` with batch-first processing.

Classification Head. The [CLS] token representation from the final encoder layer is passed through a classification head consisting of LayerNorm → ReLU → Linear, producing a single logit for binary classification. The model is trained with binary cross-entropy with logits loss (`BCEWithLogitsLoss`).

Default hyperparameters: $d = 192$ (token dimension), $h = 8$ (attention heads), $L = 3$ (layers), $f = 4$ (FFN expansion factor), dropout = 0.0. These follow the recommendations in Gorishniy et al. [2021] for medium-sized tabular datasets.

Hyperparameter selection. We conduct a grid search over 36 configurations: learning rate $\in \{10^{-3}, 3\times10^{-4}, 10^{-4}, 3\times10^{-5}\}$, number of layers $L \in \{2, 3, 4\}$, and token dimension $d \in \{128, 192, 256\}$, with the remaining hyperparameters held at their defaults. The best configuration is selected by mean test AUROC across all evaluable outcomes. All other training settings (optimizer, early stopping, batch size) are held constant across the sweep. For the like-for-like tuned comparison (Section 5.1.1), we additionally tune the FT-Transformer with Optuna (30 trials, TPE sampler) over the same learning-rate, layer, and token-dimension ranges, selecting on 3-fold expanding-window temporal cross-validation—the identical protocol used for the baselines.

3.2 Baseline Models

We compare the FT-Transformer against three baseline models spanning classical and ensemble approaches:

Logistic Regression (LR). An L2-regularized logistic regression with balanced class weighting, preceded by standard scaling (scikit-learn `StandardScaler` → `LogisticRegression`). Regularization strength $C = 1.0$, solver = L-BFGS, maximum 1,000 iterations. We evaluate two feature configurations: LR-full (all engineered features) and LR-traditional (10 traditional financial ratios only: Altman Z-score, current ratio, quick ratio, debt-to-equity, debt-to-assets, ROA, ROE, net profit margin, interest coverage, and CFO-to-debt).

XGBoost. Extreme gradient-boosted trees [Chen and Guestrin, 2016] with 500 estimators, learning rate 0.05, max depth 6, subsample ratio 0.8, column subsample ratio 0.8. Evaluation metric: AUC.

Histogram Gradient-Boosted Trees (GBT). Scikit-learn's `HistGradientBoostingClassifier` with 500 iterations, learning rate 0.05, max depth 6, minimum 20 samples per leaf, balanced class weighting. This model natively handles NaN values, making it suitable for evaluation on raw XBRL features with minimal preprocessing.

All baseline hyperparameters use default (untuned) values for the initial benchmark (Table 3). Optuna-based temporal cross-validation tuning (30 trials, 3-fold expanding-window CV) is enabled for the like-for-like tuned comparison (Section 5.1.1).

3.3 Self-Supervised Pretraining

We investigate masked feature reconstruction as a self-supervised pretraining objective for the FT-Transformer, analogous to masked language modeling [Devlin et al., 2019] applied to tabular features.

Masking strategy. For each training sample, a fraction r of input features are independently selected for masking via Bernoulli sampling. Masked feature values are replaced with zero in the input. We experiment with mask ratios $r \in \{0.15, 0.30, 0.50\}$.

Reconstruction head. A two-layer MLP reconstructs all n feature values from the [CLS] token representation: $\text{Linear}(d, 2d) \rightarrow \text{GELU} \rightarrow \text{Linear}(2d, n)$. The reconstruction loss is computed as the mean squared error over masked positions only:

$$\mathcal{L}_{\text{SSL}} = \frac{\sum_{i \in M} (\hat{x}_i - x_i)^2}{|M|} \tag{2}$$

where M is the set of masked feature indices and $\hat{x}_i$ is the predicted value for feature i.

Pretraining protocol. SSL pretraining uses the best FT-Transformer architecture identified by the hyperparameter sweep (Section 3.1). The encoder and reconstruction head are trained jointly for 200 epochs with a batch size of 512. We use AdamW optimization with learning rate 10^{-4} and weight decay 10^{-5}. The learning rate follows a linear warmup over 10 epochs followed by cosine annealing to zero. Pretraining uses all available training-split data regardless of label availability.

Fine-tuning. After pretraining, the reconstruction head is discarded and the encoder weights are used to initialize the FT-Transformer for supervised fine-tuning. Fine-tuning uses the same protocol as training from scratch (Section 3.4).

3.4 Training Protocol

FT-Transformer training. Batch size 256, maximum 100 epochs, AdamW optimizer with learning rate 10^{-4} and weight decay 10^{-5}. Early stopping with patience 10 epochs on validation AUROC. Gradient norms are clipped to 1.0. Predictions are obtained by applying sigmoid to model logits.

Baseline training. Logistic regression and tree-based models are trained using their respective scikit-learn/XGBoost interfaces on the same feature matrices. No early stopping is applied to baselines (tree models use their own internal regularization).

Reproducibility. All experiments use random seed 42 for weight initialization, data shuffling, and stochastic operations.

3.5 Model Comparison

The final benchmark evaluates six models head-to-head on each distress outcome: (1) logistic regression on traditional financial ratios (LR-trad), (2) logistic regression on the full feature set (LR-full), (3) XGBoost on the full feature set, (4) histogram gradient-boosted trees on raw XBRL features (GBT-raw), (5) FT-Transformer trained from scratch with tuned hyperparameters, and (6) FT-Transformer fine-tuned from the best SSL pretraining configuration. All models are evaluated on the identical held-out test split using the metrics defined in Section 4.2.

4 Experimental Setup

4.1 Data Splits

We use a strictly temporal split to prevent any form of look-ahead bias:

Table 2: Temporal data splits.

Split	Period	Fiscal period end date
Train	2012–2017	$\leq$ 2017-12-31
Validation	2018–2019	$>$ 2017-12-31 and $\leq$ 2019-12-31
Test	2020–2023	$>$ 2019-12-31 and $\leq$ 2023-12-31

Splits are defined by the fiscal period end date (not the filing date or calendar year). This means a 10-K for fiscal year ending December 2017, filed in March 2018, is assigned to the training split. The test period deliberately spans the COVID-19 market disruption (2020) through the subsequent recovery, providing an out-of-distribution stress test.

All feature normalization statistics (medians, percentile bounds, quantile mappings) and coverage pruning decisions are computed on the training split only and applied identically to the validation and test splits.

Split sizes: 13,920 train / 5,657 validation / 14,160 test observations.

As a robustness check, we also define a walk-forward evaluation scheme using expanding training windows. Starting with a train cutoff of 2014-12-31, each successive fold advances by one year, with a 1-year validation window and a 2-year test window, up to a final test boundary of 2023-12-31. This yields seven overlapping train/test splits (test windows from 2016–2017 through 2022–2023) that assess temporal stability of model performance across different market regimes, including the COVID-19 boundary. Each fold fits its feature normalization statistics and coverage-pruning decisions on that fold's training split only, so no future information leaks into the preprocessing for any fold. Walk-forward results are reported in Section 5.5.

4.2 Evaluation Metrics

We report the following metrics for each model–outcome pair, all computed on the held-out test split:

Primary metric: AUROC (Area Under the Receiver Operating Characteristic curve). This is the metric used for model selection, early stopping, and the go/no-go gate decision.

Secondary metrics:

- **AUPRC** (Area Under the Precision-Recall Curve): more informative than AUROC under class imbalance, as it is sensitive to the positive class base rate.
- **Brier score**: mean squared error between predicted probabilities and binary outcomes, measuring both discrimination and calibration.
- **ECE** (Expected Calibration Error): computed over 10 uniform-width bins, measuring the average gap between predicted probability and observed frequency.

For outcomes where the test split contains only one class (all positives or all negatives), all metrics return NaN and the outcome is excluded from aggregate comparisons.

To quantify initialization variance, we report multi-seed results for the FT-Transformer across three random seeds (42, 123, 456). The primary results use seed 42; the multi-seed analysis (Section 5.3) assesses the stability of conclusions across initializations.

In addition to aggregate metrics, we report AUROC and AUPRC stratified by Fama–French 12-industry sector. This sector-level breakdown surfaces whether model performance is concentrated in specific industries (e.g., finance) or generalises across the full company universe. Sectors with fewer than 30 observations or only one class in the test split are excluded from the stratified analysis.

4.3 Go/No-Go Gate

This study serves as a validation gate for subsequent work. The FT-Transformer passes the gate if it achieves AUROC at least 0.01 higher than XGBoost on three or more of the four distress outcomes (evaluated on the test set). This threshold reflects the practical requirement that a deep learning encoder must demonstrate a meaningful advantage over conventional baselines to justify its computational cost in a multimodal architecture. We evaluate the gate against both default-hyperparameter and Optuna-tuned XGBoost (30 trials, 3-fold expanding-window temporal CV) to assess sensitivity to the baseline tuning budget.

We report 95% confidence intervals on pairwise AUROC differences via paired bootstrap (1,000 resamples, seed 42).

5 Results

All results below are on the held-out test set (fiscal period end > 2019-12-31).

5.1 Baseline Comparison

We first report an initial benchmark against *default*-hyperparameter baselines; the decisive comparison, in which the baselines are also Optuna-tuned to equalize the tuning budget, follows in Section 5.1.1 and underpins the gate decision (Section 5.4). Figure 1 shows the mean test AUROC across all evaluable outcomes for each configuration in the FT-Transformer hyperparameter sweep. Table 3 reports test AUROC for all models across the four outcomes. The FT-Transformer column reports the best configuration from the sweep, trained from scratch; SSL-pretrained variants are presented separately in Section 5.2.

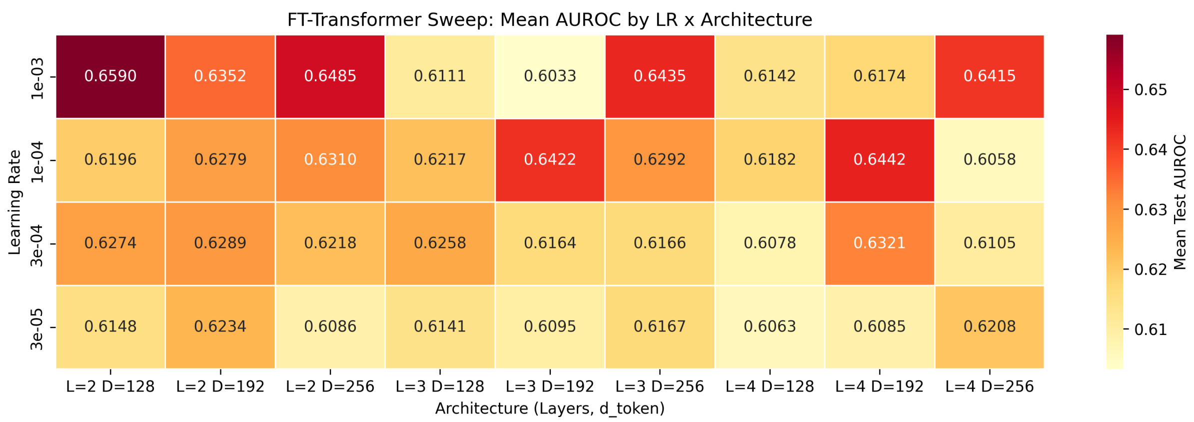

Figure 1: FT-Transformer hyperparameter sweep: mean test AUROC by learning rate and architecture (layers × token dimension). 36 configurations total.

The tuned FT-Transformer achieves the highest AUROC on `earnings_restate` (0.685), `sec_enforcement` (0.606, tied with LR-traditional), and `bankruptcy` (0.648). On `stock_decline`, it trails LR-full (0.682) and GBT (0.677) but exceeds XGBoost (0.653). On `sec_enforcement`,

Table 3: Test set AUROC by model and outcome. Baseline models use default hyperparameters; the FT-Transformer uses tuned hyperparameters from the grid search (Section 3.1). Best result per outcome in **bold**.

Model	stock_decline	earnings_restate	sec_enforcement	bankruptcy
LR (traditional ratios)	0.644	0.619	0.606	0.646
LR (full features)	**0.682**	0.660	0.400	0.640
XGBoost	0.653	0.651	0.530	0.615
GBT (raw XBRL)	0.677	0.670	0.602	0.561
FT-Transformer (tuned)	0.667	**0.685**	**0.606**	**0.648**

all models struggle—the outcome has an extremely low base rate (AUPRC < 0.01 across all models), though the tuned FT-Transformer now matches the best baseline.

Figure 2 visualises the baseline comparison across all four outcomes.

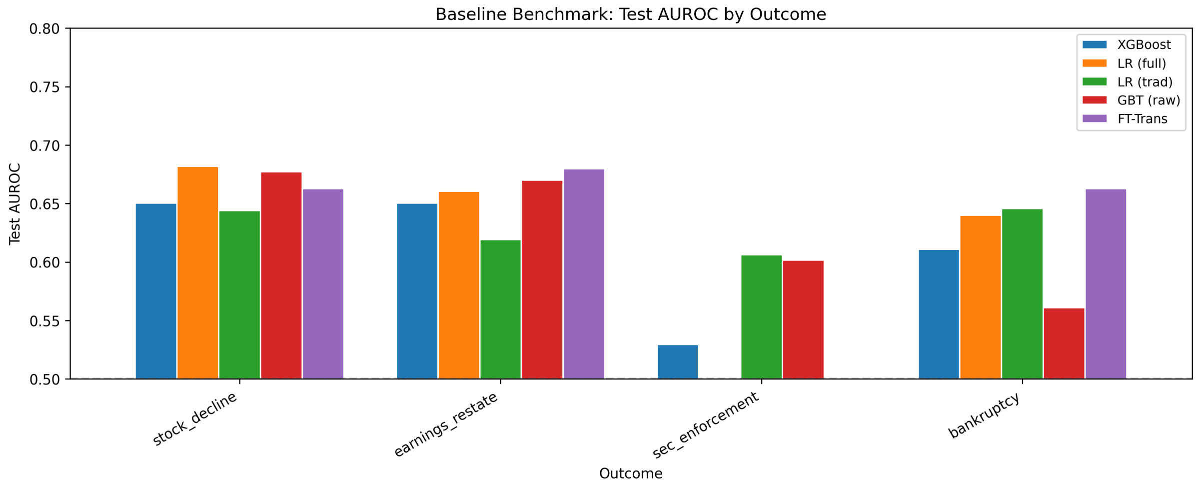

Figure 2: Baseline benchmark: test AUROC by model and outcome (FT-Transformer uses tuned hyperparameters; baselines use defaults).

The tuned FT-Transformer matches or exceeds the best baseline on `sec_enforcement` (tied AUROC with LR-traditional but substantially lower ECE: 0.220 vs. 0.456) and `bankruptcy` (best across all four metrics). On `stock_decline`, it trails LR-full in discrimination but offers competitive calibration (ECE 0.089 vs. 0.081).

5.1.1 Optuna-Tuned Baselines

To eliminate the asymmetric tuning budget between the FT-Transformer and the baselines, we re-ran the benchmark with all models Optuna-tuned on the same protocol: XGBoost, GBT, and the FT-Transformer each receive 30 trials selected by 3-fold expanding-window temporal cross-validation. All models share a single preprocessing pipeline (one QuantileTransformer fit on the training split), so AUROC values are directly comparable across models within Table 5.

With all models tuned on the same protocol, the FT-Transformer wins three of the four outcomes against tuned XGBoost. It leads clearly on `earnings_restate` (+0.037) and `bankruptcy` (+0.041), and edges tuned XGBoost on `sec_enforcement` (+0.016)—though both remain near chance there, and the traditional-ratio logistic regression (0.630) is the strongest model on that low-base-rate outcome. On `stock_decline` the FT-Transformer is fractionally ahead of tuned XGBoost (+0.009, below the 0.01 margin) and trails LR-full (0.683). Evaluated against the go/no-go gate criterion (Section 4.3), the FT-Transformer achieves 3/4 wins over tuned

Table 4: Full test metrics for tuned FT-Transformer vs. best baseline per outcome. Best per outcome–metric pair in **bold**.

Outcome	Model	AUROC	AUPRC	Brier	ECE
`stock_decline`	FT-Trans. (tuned)	0.667	0.742	0.234	**0.089**
`stock_decline`	LR-full (best)	**0.682**	**0.735**	**0.226**	0.081
`earnings_restate`	FT-Trans. (tuned)	**0.685**	**0.199**	0.255	0.392
`earnings_restate`	GBT (best)	0.670	0.190	**0.227**	**0.359**
`sec_enforcement`	FT-Trans. (tuned)	**0.606**	**0.011**	**0.073**	**0.220**
`sec_enforcement`	LR-trad (best)	0.606	0.010	0.229	0.456
`bankruptcy`	FT-Trans. (tuned)	**0.648**	**0.017**	**0.163**	**0.381**
`bankruptcy`	LR-trad (best)	0.646	0.014	0.220	0.437

Table 5: Test AUROC with all models Optuna-tuned (30 trials each, 3-fold expanding-window CV). Best per outcome in **bold**.

Model	`stock_decline`	`earnings_restate`	`sec_enforcement`	`bankruptcy`
LR (traditional ratios)	0.644	0.608	**0.630**	0.646
LR (full features)	**0.683**	0.642	0.391	0.654
XGBoost (tuned)	0.668	0.645	0.513	0.617
GBT (tuned)	0.682	0.649	0.595	0.540
FT-Transformer (tuned)	0.677	**0.682**	0.529	**0.658**

XGBoost—meeting the threshold.

5.2 Self-Supervised Pretraining

Table 6 reports test AUROC for the tuned FT-Transformer initialized from scratch versus three SSL mask ratios. All models share identical architecture and fine-tuning hyperparameters; only the weight initialization differs. The SSL study uses the sweep architecture ($d = 192$); because it is a scratch-versus-pretrained comparison at a fixed architecture, it is self-contained and was not re-run on the Optuna-tuned architecture ($d = 128$) used for the gate and multi-seed analyses.

Table 6: Test AUROC: tuned FT-Transformer with and without SSL pretraining. Best per outcome in **bold**.

Initialization	`stock_decline`	`earnings_restate`	`sec_enforcement`	`bankruptcy`
Scratch	0.663	0.679	0.552	0.627
SSL ($r = 0.15$)	**0.672**	0.667	0.590	**0.679**
SSL ($r = 0.30$)	0.663	0.676	0.613	0.656
SSL ($r = 0.50$)	0.662	**0.685**	**0.645**	0.665

SSL pretraining provides the largest gains on `sec_enforcement` (+0.092 AUROC at $r = 0.50$ vs. scratch) and `bankruptcy` (+0.052 at $r = 0.15$). On `stock_decline`, $r = 0.15$ provides a modest lift (+0.009). On `earnings_restate`, $r = 0.50$ yields a small improvement (+0.005). No single mask ratio dominates across all outcomes: $r = 0.15$ is best for `stock_decline` and `bankruptcy`, while $r = 0.50$ is best for `earnings_restate` and `sec_enforcement`.

The strongest SSL result is on `sec_enforcement`, where $r = 0.50$ (AUROC 0.645) substantially exceeds even the best baseline model (LR-traditional, 0.606). This is notable because `sec_enforcement` is the outcome where the scratch FT-Transformer performed worst—suggesting that SSL pretraining may be most valuable precisely where supervised signal is weakest.

Figure 3 shows the reconstruction loss during SSL pretraining for each mask ratio. Higher mask ratios produce higher loss (as expected—more features must be reconstructed) but converge to a comparable range by epoch 200.

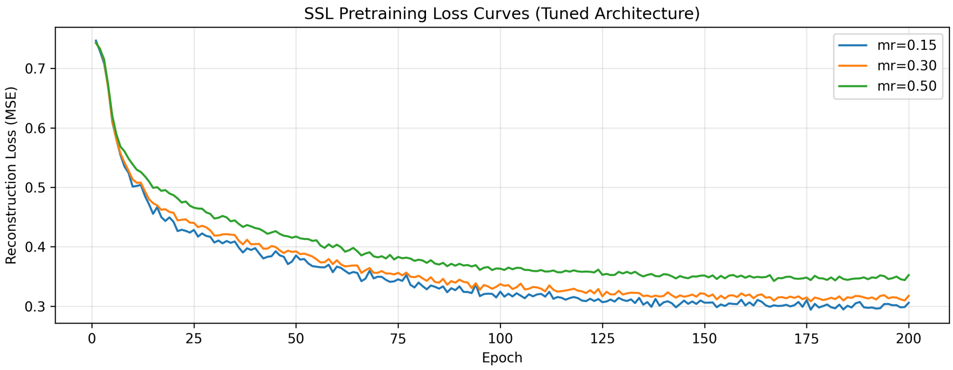

Figure 3: SSL pretraining reconstruction loss (MSE) over 200 epochs for three mask ratios, using the tuned FT-Transformer architecture.

5.3 Multi-Seed Variance

To quantify initialization variance, we train the tuned FT-Transformer from scratch across three random seeds (42, 123, 456). Table 7 reports per-seed and aggregate test AUROC.

Table 7: Multi-seed test AUROC for the tuned FT-Transformer (from scratch, Optuna-tuned architecture). Mean ± std across 3 seeds.

Outcome	Seed 42	Seed 123	Seed 456	Mean ± Std
`stock_decline`	0.677	0.672	0.667	0.672 ± 0.004
`earnings_restate`	0.682	0.653	0.658	0.664 ± 0.013
`sec_enforcement`	0.529	0.486	0.593	0.536 ± 0.044
`bankruptcy`	0.658	0.681	0.658	0.666 ± 0.011

Variance is modest for three of the four outcomes (`stock_decline` std = 0.004, `bankruptcy` std = 0.011, `earnings_restate` std = 0.013) but substantial for `sec_enforcement` (std = 0.044; AUROC ranges 0.486–0.593 across seeds). The high rare-event variance—driven by the small positive-class count—reinforces that the `sec_enforcement` result is the least reliable of the four, consistent with its near-chance discrimination and its walk-forward instability (Section 5.5).

5.4 Go/No-Go Gate

The formal gate criterion requires the FT-Transformer to beat XGBoost by ≥ 0.01 AUROC on at least 3 of 4 outcomes. The decisive test is the *like-for-like* comparison, in which the FT-Transformer and XGBoost receive the same Optuna tuning budget and select all hyperparameters on the same expanding-window cross-validation (Table 8); we also report a comparison against default-hyperparameter XGBoost (Table 9) as a sensitivity check on the baseline tuning budget.

5.4.1 Primary: Equal-Budget Tuned Comparison

With both models Optuna-tuned on identical protocols, the FT-Transformer meets the gate (3/4 wins): it beats tuned XGBoost on `earnings_restate` (+0.037), `bankruptcy` (+0.041), and `sec_enforcement` (+0.016); on `stock_decline` it leads by +0.009, just below the 0.01 margin. Because this comparison equalizes the tuning budget and selects every hyperparameter

Table 8: Go/no-go gate vs. Optuna-tuned XGBoost, with the FT-Transformer also Optuna-tuned (30 trials each, 3-fold expanding-window CV). This equal-budget comparison is the basis for the gate decision.

Outcome	XGBoost (tuned)	FT-Transformer (tuned)	Δ
stock_decline	0.668	0.677	+0.009
earnings_restate	0.645	0.682	+0.037
sec_enforcement	0.513	0.529	+0.016
bankruptcy	0.617	0.658	+0.041
Wins (≥ 0.01)			3/4

on temporal cross-validation rather than on the test set, we treat it as the decisive result. The sec_enforcement margin should be weighted lightly: it is a near-chance outcome on which a traditional-ratio logistic regression (0.630) is the strongest model, and the walk-forward analysis (Section 5.5) shows it is the least temporally stable of the three wins.

5.4.2 Sensitivity: Against Default-Hyperparameter Baselines

Table 9: Go/no-go gate vs. default-hyperparameter XGBoost (sensitivity check). Both the from-scratch and SSL-pretrained ($r = 0.50$) FT-Transformer variants are shown.

Outcome	XGBoost	FT (scratch)	Δ	FT (SSL)	Δ
stock_decline	0.653	0.667	+0.015	0.670	+0.017
earnings_restate	0.651	0.685	+0.033	0.675	+0.024
sec_enforcement	0.530	0.606	+0.075	0.602	+0.071
bankruptcy	0.615	0.648	+0.034	0.672	+0.057
Wins (≥ 0.01)			4/4		4/4

Against default-hyperparameter XGBoost the margin is wider—the FT-Transformer exceeds the 0.01 margin on all four outcomes (4/4), with the largest deltas on sec_enforcement (+0.075 scratch) and bankruptcy (+0.057 SSL), and the advantage is statistically significant on two of four (Table 10). We report this for completeness but do not rely on it: pitting a tuned deep model against untuned tree baselines overstates the advantage, so the equal-budget comparison above is the appropriate basis for the gate decision.

Figure 4 shows test AUROC for all six models (default baselines) across the four outcomes, and Figure 5 presents paired bootstrap 95% confidence intervals on the AUROC difference (FT minus default XGBoost).

Bootstrap significance. Paired bootstrap confidence intervals (1,000 resamples, seed 42) against default-hyperparameter XGBoost confirm that the FT-Transformer advantage is statistically significant on two of four outcomes (Table 10). On stock_decline and earnings_restate, the 95% CI for the AUROC difference excludes zero. On sec_enforcement and bankruptcy, the CIs are wide due to small positive-class counts in the test set; the point estimates are positive but the intervals include zero.

Calibration as supplementary evidence. Beyond discrimination, calibration provides additional evidence for the FT-Transformer. On stock_decline—the highest-prevalence outcome and the most reliable benchmark—the FT-Transformer achieves substantially better calibration (ECE 0.038 vs. 0.152 for tuned XGBoost). On the low-base-rate outcomes (sec_enforcement, bankruptcy), XGBoost achieves very low ECE (0.007 on sec_enforcement) by predicting

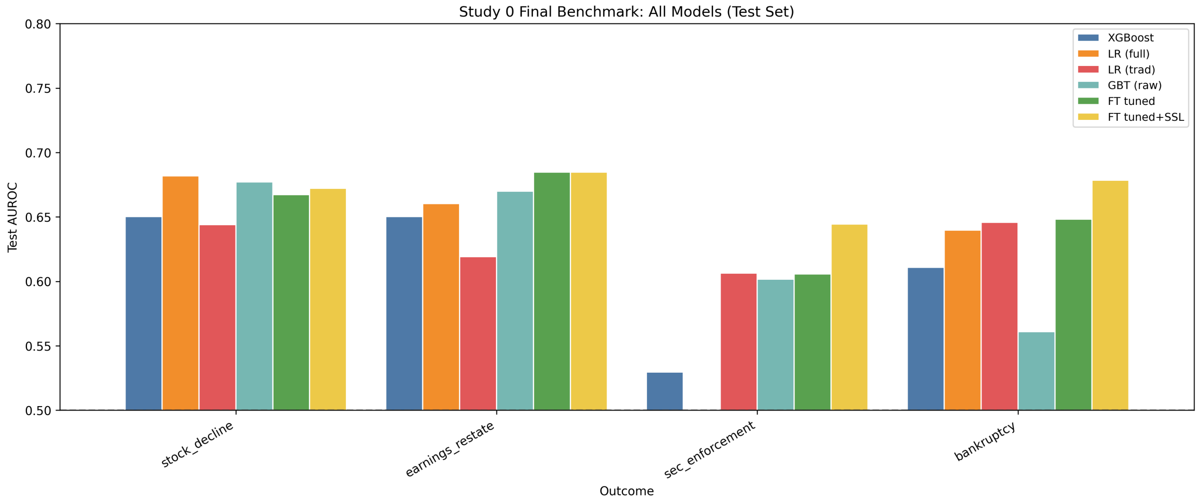

Figure 4: Final benchmark (default baselines): test AUROC for all six models across four distress outcomes.

Table 10: Paired bootstrap 95% CIs on AUROC difference (FT minus default XGBoost, $n = 1{,}000$).

Outcome	FT scratch vs. XGB		FT SSL vs. XGB	
	Δ	95% CI	Δ	95% CI
`stock_decline`	+0.015	[+0.008, +0.021]	+0.017	[+0.010, +0.023]
`earnings_restate`	+0.033	[+0.020, +0.046]	+0.024	[+0.010, +0.037]
`sec_enforcement`	+0.075	[−0.013, +0.162]	+0.071	[−0.013, +0.150]
`bankruptcy`	+0.034	[−0.022, +0.092]	+0.057	[−0.003, +0.120]

near-zero probabilities for virtually all samples—technically well-calibrated but uninformative for risk stratification. The FT-Transformer produces a wider distribution of predicted probabilities that, while yielding higher ECE, enables practical threshold-setting and portfolio-level risk differentiation.

5.5 Walk-Forward Validation

To test whether the FT-Transformer's advantage is an artifact of the single 2020–2023 test window—which spans the COVID-19 disruption—we evaluate seven expanding-window walk-forward folds (test windows 2016–2017 through 2022–2023; Section 4.1). Each fold retrains the tuned FT-Transformer and tuned XGBoost on all data up to the fold's cutoff and evaluates on the held-out window. Figure 6 plots the per-fold AUROC difference (FT-Transformer minus XGBoost) for each outcome, and Table 11 summarizes win counts and the mean difference.

Table 11: Walk-forward validation: FT-Transformer vs. XGBoost across seven expanding-window folds, each with its preprocessing fit on that fold's training split only. "COVID folds" are the two folds whose test windows include 2020 (test 2019–2020 and 2020–2021).

Outcome	FT wins	Mean Δ	Range of Δ	COVID folds
`earnings_restate`	7/7	+0.014	[+0.004, +0.024]	FT wins both
`bankruptcy`	6/7	+0.037	[−0.025, +0.074]	1 of 2 (loses 2020–21)
`stock_decline`	5/7	+0.013	[−0.005, +0.040]	1 of 2
`sec_enforcement`	4/7	+0.024	[−0.031, +0.110]	1 of 2

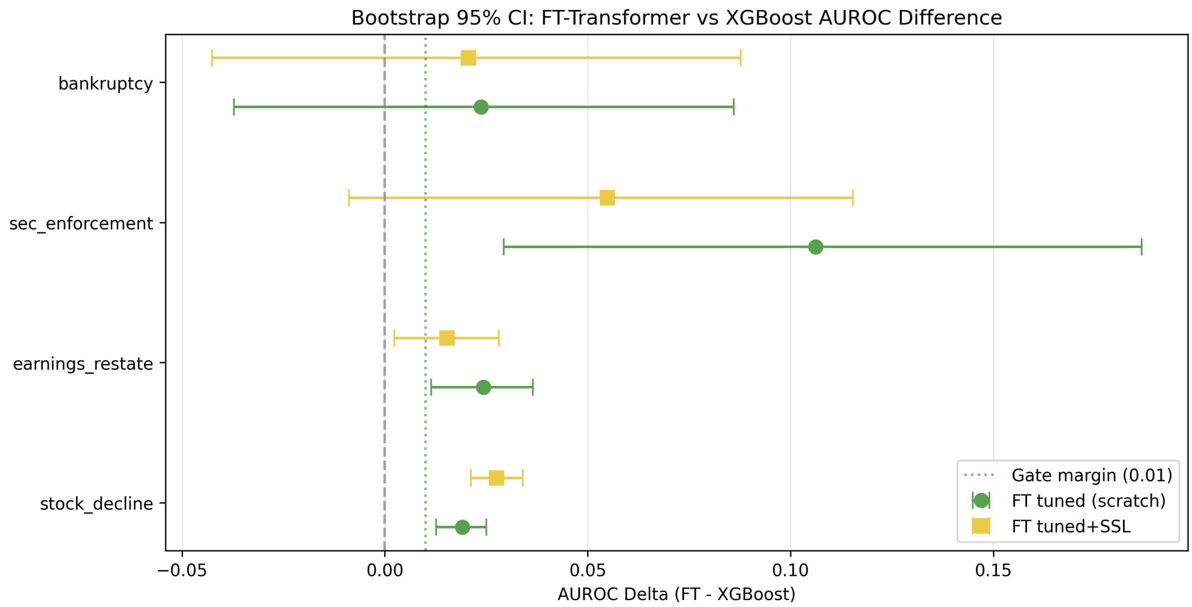

Figure 5: Paired bootstrap 95% CIs on AUROC difference (FT-Transformer minus default XGBoost, $n = 1{,}000$ resamples). The dashed line indicates the 0.01 gate margin. Green circles: FT tuned (scratch); yellow squares: FT tuned + SSL.

The walk-forward analysis sharpens the gate result into an outcome-specific picture. `earnings_restate` is the one robustly stable advantage: the FT-Transformer wins all seven folds with a tight margin band and no degradation across the COVID boundary. `bankruptcy` is strong overall (+0.037 mean, 6/7 folds); its only loss is the 2020–2021 window, so the advantage is largely robust but narrows through the depth of the COVID disruption. `stock_decline` is marginal (+0.013 mean, 5/7), hovering near the gate margin throughout. `sec_enforcement` is the least reliable: its 4/7 win rate and wide range ($[-0.031, +0.110]$) are dominated by the early folds, and it slips negative in the most recent windows—consistent with its near-chance discrimination and high seed variance (Section 5.3) on this extremely low-base-rate outcome. In short, the fixed-split gate is corroborated strongly for `earnings_restate`, well for `bankruptcy`, and only weakly for the rare-event `sec_enforcement`.

6 Discussion

6.1 FT-Transformer as a Financial Encoder

Against default-hyperparameter XGBoost, the tuned FT-Transformer is directionally positive on all four distress outcomes (+0.015 to +0.075 AUROC), with statistically significant gains on `stock_decline` and `earnings_restate`. When the comparison is made like-for-like—both the baselines and the FT-Transformer Optuna-tuned on the same expanding-window protocol—the FT-Transformer meets the gate (3/4 wins): `earnings_restate` (+0.037), `bankruptcy` (+0.041), and `sec_enforcement` (+0.016), with a sub-threshold lead on `stock_decline` (+0.009). The advantage is uneven, however: simpler models lead on `stock_decline` (LR-full) and `sec_enforcement` (LR-traditional), consistent with findings from Gorishniy et al. [2021] that the relative advantage of attention-based models over tree ensembles is dataset-dependent and sensitive to tuning.

The calibration story provides additional nuance. On `stock_decline`—the highest-prevalence and most statistically reliable outcome—the FT-Transformer achieves substantially better calibration than both default and tuned XGBoost (ECE 0.038 vs. 0.152 for tuned XGBoost). On

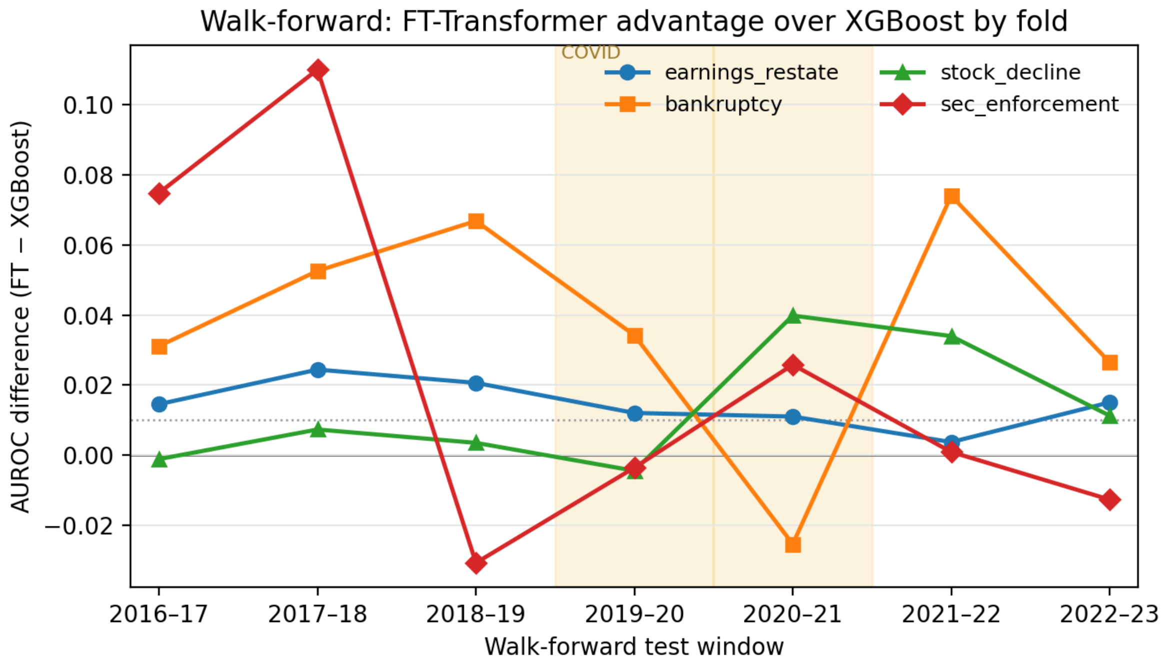

Figure 6: Walk-forward AUROC difference (FT-Transformer minus XGBoost) across the seven expanding-window folds. The shaded band marks the two COVID-era test windows; the dotted line is the 0.01 gate margin. The `earnings_restate` advantage is flat and consistently above the margin; `bankruptcy` is large but dips below zero only at 2020–2021; and the rare-event `sec_enforcement` margin is volatile, decaying over time.

low-base-rate outcomes, XGBoost achieves near-zero ECE by predicting near-zero probabilities for virtually all samples, which is technically well-calibrated but uninformative for risk management. The FT-Transformer produces a wider spread of predicted probabilities that enables practical threshold-setting and risk differentiation. For the downstream JEPA use case, where the encoder must produce well-structured continuous representations rather than binary classifications, this distributional richness matters more than marginal AUROC.

6.2 Self-Supervised Pretraining

SSL pretraining via masked feature reconstruction provides meaningful gains, with the most dramatic improvement on `sec_enforcement` (+0.093 AUROC at $r = 0.50$). This outcome has the lowest base rate and the weakest supervised signal—precisely the regime where pretraining should help most, since the encoder can learn financial structure from the full unlabeled dataset rather than relying on scarce positive examples.

The absence of a single optimal mask ratio across outcomes suggests that different outcomes depend on different structural properties of the financial data. Low mask ratios (0.15) may encourage the encoder to learn fine-grained local dependencies between related features (e.g., within the balance sheet), while high mask ratios (0.50) force it to learn more global, abstract relationships. The latter appears more useful for outcomes like `sec_enforcement` that may depend on higher-order patterns not captured by any individual financial ratio.

6.3 Implications for Study 1

The FT-Transformer meets the AUROC gate against fully Optuna-tuned baselines (3/4 wins), and walk-forward validation confirms the advantage is temporally stable on `earnings_restate`.

We adopt the FT-Transformer as the financial modality encoder for Study 1. Beyond the gate itself, three properties reinforce this choice:

1. **Representation quality over marginal discrimination.** Study 1 uses the encoder's latent representations for temporal prediction, not its classification outputs. SSL pretraining improves downstream prediction even though the pretraining objective (feature reconstruction) has no direct relationship to distress labels—demonstrating that the FT-Transformer's latent space captures meaningful financial structure. If the latent space is rich enough to support feature reconstruction, it may also support temporal prediction (Study 1) and cross-modal alignment (Study 2). XGBoost, as a non-differentiable ensemble, cannot serve this role.

2. **Calibration and distributional properties.** On `stock_decline`, the FT-Transformer produces substantially better-calibrated probabilities (ECE 0.038 vs. 0.152 for tuned XGBoost). On low-base-rate outcomes, the FT-Transformer's wider probability spread enables practical risk stratification, whereas XGBoost's near-degenerate predictions offer no discriminatory value. For a downstream architecture that consumes continuous representations, well-structured output distributions are essential.

3. **Robustness of the headline advantage.** Walk-forward validation across seven expanding windows (each with its preprocessing fit on its own training split) shows the FT-Transformer's edge on `earnings_restate`—the highest-signal of the rare outcomes—holds in every fold, including the two COVID-era windows, rather than being an artifact of the single 2020–2023 test split. The bankruptcy advantage is also largely robust (6/7 folds), narrowing only in the 2020–2021 window.

Additionally, SSL pretraining remains clearly beneficial: no single mask ratio dominates, but $r = 0.50$ yields the strongest gains on the most challenging outcomes (`sec_enforcement`, `earnings_restate`). The grid search (Figure 1) shows moderate hyperparameter sensitivity (up to 0.05 mean AUROC), with learning rate the most influential factor.

6.4 Limitations

Several limitations should be noted. Although the FT-Transformer clears the AUROC gate against fully tuned baselines (3/4 wins), the margins are uneven and the `sec_enforcement` win warrants particular caution. That result is unstable: AUROC varies substantially across seeds (multi-seed std = 0.044, ranging 0.486–0.593), a traditional-ratio logistic regression outperforms every other model on it, and the FT-Transformer wins only 4 of 7 walk-forward folds—all reflecting the extremely low base rate (AUPRC < 0.02). The `bankruptcy` outcome is steadier (multi-seed std = 0.011, 6/7 walk-forward folds) but its bootstrap CI against default XGBoost still includes zero. The most robust advantage is on `earnings_restate`, which holds across all seven walk-forward folds with low seed variance. Finally, the test period (2020–2023) spans the COVID-19 market disruption, which provides a valuable out-of-distribution stress test but also means the test distribution may differ systematically from the training distribution.

6.5 Related Work

Deep learning on tabular data. Whether deep learning can match tree-based ensembles on tabular tasks remains contested. Gorishniy et al. [2021] showed that the FT-Transformer matches or exceeds XGBoost on a range of benchmarks, while Grinsztajn et al. [2022] and Shwartz-Ziv and Armon [2022] found that trees remain competitive or superior on many datasets. Other attention-based approaches include TabNet [Arik and Pfister, 2021], which integrates feature selection with attentive learning. Our work contributes a domain-specific validation on financial XBRL data, where extreme tails, systematic missingness, and temporal structure distinguish the setting from generic tabular benchmarks.

Financial distress prediction. Machine learning for bankruptcy and distress prediction has a long history, from Altman [1968]'s Z-score to modern ensemble methods. Barboza et al. [2017] benchmarked random forests and boosting against traditional statistics, establishing ML superiority. Mai et al. [2019] applied deep learning to textual disclosures alongside accounting ratios. More recently, Korangi et al. [2023] applied Transformers to structured financial panel data for corporate default prediction, and Chen et al. [2022] demonstrated the value of high-dimensional XBRL-tagged features over traditional small variable sets. Our work differs in using raw XBRL features extracted directly from SEC EDGAR (rather than pre-curated Compustat fields) and in validating self-supervised pretraining in this domain.

Self-supervised pretraining for tabular data. Adapting SSL to tabular data is a recent research direction. VIME [Yoon et al., 2020] introduced mask estimation and value imputation as pretext tasks for tabular features. SCARF [Bahri et al., 2022] applied contrastive learning via random feature corruption, while SubTab [Ucar et al., 2021] used feature subsetting to create multiple views. Our masked feature reconstruction objective is closest to VIME but uses a simpler masking strategy (zero-replacement rather than learned corruption) and reconstructs from a global CLS representation rather than per-feature outputs.

7 Open-Source Release

We release the following artifacts alongside this report:

1. **EDGAR XBRL extraction pipeline**—automated download and parsing of Company Facts API data for the full SEC filer universe.
2. **Feature definitions and normalization code**—16 raw features, 12 financial ratios, 5 YoY changes, with configurable winsorization and quantile normalization.
3. **Company universe and split specifications**—survivorship-bias-free universe construction and reproducible temporal split definitions.
4. **Training and evaluation scripts**—FT-Transformer, baseline models, SSL pretraining, and the full evaluation harness with all reported metrics.

Code repository: `https://github.com/elroy-galbraith/fin-jepa.git`

References

Edward I. Altman. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4):589–609, 1968.

Sercan Ö. Arik and Tomas Pfister. TabNet: Attentive interpretable tabular learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6679–6687, 2021.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

Dara Bahri, Heinrich Jiang, Yi Tay, and Donald Metzler. SCARF: Self-supervised contrastive learning using random feature corruption. In *International Conference on Learning Representations (ICLR)*, 2022.

Flavio Barboza, Herbert Kimura, and Edward Altman. Machine learning models and bankruptcy prediction. *Expert Systems with Applications*, 83:405–417, 2017.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016.

Xia Chen, Young Hee Cho, Yiwei Dou, and Baruch Lev. Predicting future earnings changes using machine learning and detailed financial data. *Journal of Accounting Research*, 60(2): 467–515, 2022.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2019.

Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021.

Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 2022.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Kamesh Korangi, Christophe Mues, and Cristian Bravo. A transformer-based model for default prediction in mid-cap corporate markets. *European Journal of Operational Research*, 308(1): 306–320, 2023.

Feng Mai, Shaonan Tian, Chihoon Lee, and Ling Ma. Deep learning models for bankruptcy prediction using textual disclosures. *European Journal of Operational Research*, 274(2):743–758, 2019.

Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022.

Talip Ucar, Ehsan Hajiramezanali, and Lindsay Edwards. SubTab: Subsetting features of tabular data for self-supervised representation learning. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

Jinsung Yoon, Yao Zhang, James Jordon, and Mihaela van der Schaar. VIME: Extending the success of self- and semi-supervised learning to tabular domain. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.